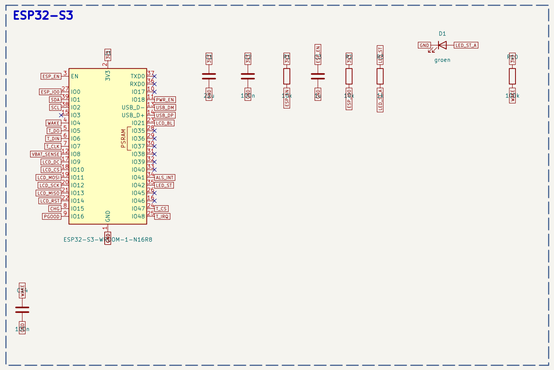
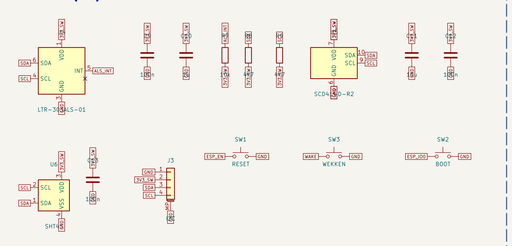
ESP32-S3



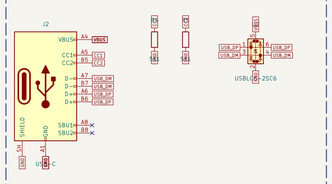
SENSOREN (I2C)



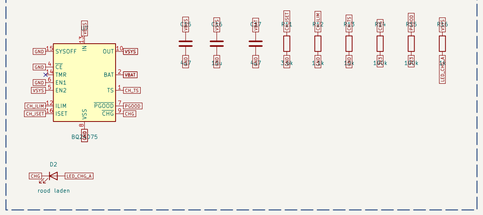
BEVESTIGING



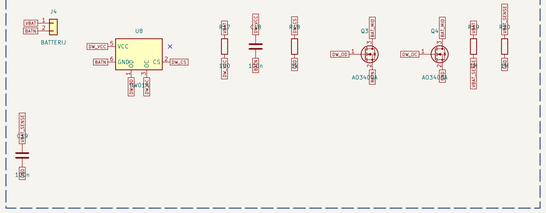
USB-C



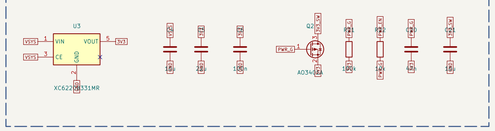
LADEN (BQ24075)



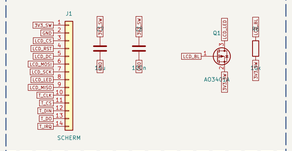
BATTERIJ + BEVEILIGING



3,3 V + SCHAKELBARE VOEDING



SCHERM 2.8" ILI9341 + TOUCH



LUCHTMETER S3 v1.0 — ESP32-S3 air-quality meter with 2.8in touchscreen, battery powered (generated by tools/build_sch.py from tools/design.py)
Sensors on I2C (SDA IO1, SCL IO2): SCD41 CO2 0x62, SHT45 temp/RH 0x44, LTR-303ALS light 0x29 (INT IO41). Qwiic for extra sensors.
Display MSP2807 (ILI9341): SCK IO12, MOSI IO11, MISO IO13, CS IO10, DC IO9, RST IO14, backlight IO21 (active low). Touch XPT2046: CLK IO7, DIN IO6, DO IO5, CS IO47, IRQ IO48.
Power: USB-C -> BQ24075 (charge ~0.9 A, power path) -> 18650 on BAT+/BAT- (DW01A protection) -> XC6220 3.3 V. 3V3_SW (display + sensors) on when PWR_EN (IO18) is low.
Deep sleep: set PWR_EN high (rail off), set all display/I2C pins to input, wake on timer or the WEKKEN button (IO4, ext0). Battery: VBAT = ADC(IO8) x 2. CHG IO15, PGOOD IO16 (low = USB).

ESP32-S3 + SCD41 + SHT45 + LTR-303 + 2.8in touch, 18650
Kobe Van Robaeys — maakleerplek
Sheet: /
File: luchtmeter.kicad_sch
Title: Luchtmeter S3
Size: A1 | Date: 2026-10-18 | **Rev: 1.0**
KiCad E.D.A. 10.0.3 | Id: 1/1